

Online Resource 1

Supplementary tables, figure and questionnaire for: Internal Control Architectures and Accountability Dimensions in Emerging Economy Bureaucracies

S1 Sample and measurement

Table S1 Demographic characteristics of the respondents (N = 187)

Variable	Category	Frequency	Percentage (%)	Cumulative (%)
Gender	Male	126	67.4	67.4
	Female	61	32.6	100.0
Age	20–25 years	19	10.2	10.2
	26–30 years	53	28.3	38.5
	31–35 years	50	26.7	65.2
	Over 35 years	65	34.8	100.0
Level of Education	Diploma	1	0.5	0.5
	Bachelor's Degree	18	9.6	10.2
	Master's Degree	92	49.2	59.4
	Ph.D.	76	40.6	100.0
Professional Experience	Less than 5 years	41	21.9	21.9
	6–10 years	72	38.5	60.4
	11–15 years	43	23.0	83.4
	Over 15 years	31	16.6	100.0

Note. Source: survey data.

Table S2 Item statistics and CFA loadings (N = 187)

Item	Construct	Mean	SD	Skewness	Kurtosis	λ	b	SE	t	λ^2
Q1	CA	3.481	0.986	-0.543	0.076	0.754	1.000	–	–	0.569
Q2	CA	3.401	1.050	-0.243	-0.546	0.755	1.066	0.100	10.692	0.570
Q3	CA	3.460	1.054	-0.479	-0.303	0.820	1.162	0.099	11.753	0.672
Q4	CA	3.455	1.074	-0.355	-0.514	0.790	1.141	0.101	11.266	0.624
Q5	CA	3.417	0.999	-0.326	-0.244	0.727	0.977	0.095	10.245	0.529
Q6	CA	3.647	0.986	-0.468	-0.098	0.797	1.057	0.093	11.374	0.635
Q7	CA	3.508	1.007	-0.549	-0.019	0.722	0.979	0.096	10.169	0.522
Q8	CA	3.401	0.942	-0.335	-0.213	0.796	1.008	0.089	11.357	0.633
Q9	CA	3.503	0.991	-0.309	-0.333	0.807	1.076	0.093	11.543	0.651
Q10	FM	3.701	1.019	-0.726	0.198	0.677	1.000	–	–	0.459
Q11	FM	3.674	1.008	-0.453	-0.349	0.748	1.093	0.130	8.419	0.559
Q12	FM	3.567	0.950	-0.289	-0.348	0.718	0.989	0.121	8.174	0.516
Q13	FM	3.508	0.964	-0.406	-0.160	0.678	0.947	0.121	7.810	0.460
Z1	ORG	3.882	0.960	-0.609	-0.192	0.816	1.000	–	–	0.667
Z2	ORG	3.642	1.055	-0.462	-0.263	0.856	1.153	0.081	14.153	0.733
Z3	ORG	3.567	1.102	-0.415	-0.418	0.834	1.173	0.086	13.601	0.696
Z4	ORG	3.471	1.038	-0.430	-0.064	0.830	1.099	0.081	13.493	0.688
Z5	ORG	3.695	1.036	-0.563	-0.274	0.826	1.092	0.081	13.402	0.682
Z6	ORG	3.535	1.137	-0.430	-0.635	0.873	1.267	0.087	14.587	0.763
Z7	ORG	3.877	0.995	-0.544	-0.457	0.807	1.024	0.079	12.945	0.651
Z8	LEG	3.572	1.062	-0.246	-0.750	0.684	1.000	–	–	0.468
Z9	LEG	3.358	1.095	-0.353	-0.536	0.871	1.313	0.123	10.670	0.759

Item	Construct	Mean	SD	Skewness	Kurtosis	λ	b	SE	t	λ^2
Z10	LEG	3.439	1.112	-0.223	-0.796	0.768	1.176	0.123	9.572	0.590
Z11	LEG	3.465	1.028	-0.101	-0.669	0.836	1.184	0.115	10.316	0.700
Z12	LEG	3.513	0.991	-0.506	-0.032	0.813	1.110	0.110	10.070	0.662
Z13	LEG	3.508	1.007	-0.453	-0.151	0.685	0.950	0.110	8.627	0.470
Z14	PRO	3.749	1.009	-0.431	-0.608	0.733	1.000	–	–	0.538
Z15	PRO	3.529	0.980	-0.222	-0.391	0.800	1.060	0.097	10.966	0.641
Z16	PRO	3.829	0.929	-0.587	0.129	0.774	0.972	0.092	10.584	0.600
Z17	PRO	3.701	1.003	-0.467	-0.168	0.836	1.134	0.099	11.493	0.700
Z18	PRO	3.374	1.135	-0.219	-0.827	0.874	1.342	0.111	12.040	0.764
Z19	PRO	3.583	1.046	-0.379	-0.410	0.721	1.020	0.104	9.812	0.520
Z20	PRO	3.604	1.049	-0.534	-0.346	0.797	1.130	0.104	10.910	0.635
Z21	POL	3.439	1.078	-0.139	-0.810	0.803	1.000	–	–	0.645
Z22	POL	3.422	1.020	-0.217	-0.582	0.789	0.930	0.081	11.525	0.623
Z23	POL	3.460	1.038	-0.169	-0.729	0.792	0.950	0.082	11.574	0.627
Z24	POL	3.273	1.070	-0.298	-0.358	0.834	1.032	0.084	12.306	0.696

Note. λ = standardized loading; b = unstandardized loading; SE = standard error; t = critical ratio; λ^2 = item reliability. The first loading of each construct (b = 1.000) was fixed. All free loadings are significant at $p < 0.001$. Skewness and kurtosis are bias-corrected.

S2 Distributional properties

Table S3 Univariate normality tests of the constructs

Construct	K–S D	K–S p	Lilliefors p	S–W W	S–W p	z skewness	z kurtosis
Control Activities	0.072	0.268	0.023	0.977	0.004	-2.757	0.484
Financial Management	0.077	0.203	0.011	0.973	0.001	-2.274	-0.025
Organizational	0.081	0.162	0.006	0.962	< 0.001	-2.897	-0.793
Legal	0.075	0.236	0.018	0.974	0.001	-1.097	-2.298
Professional	0.069	0.325	0.040	0.975	0.002	-2.149	-0.891
Political	0.095	0.063	0.001	0.977	0.004	-0.612	-1.896

Note. K–S = Kolmogorov–Smirnov; S–W = Shapiro–Wilk. Shapiro–Wilk rejected normality for all 37 items ($p < 0.001$).

Table S4 Multivariate normality (Mardia’s test)

Coefficient	Statistic	Expected value / df	Test value	p
Multivariate skewness (χ^2)	301.033	9,139	9,382.183	0.037
Multivariate skewness (χ^2 , small-sample corrected)	301.033	9,139	9,540.712	0.002
Multivariate kurtosis (z)	1,444.832	1,443	0.233	0.816

Note. The small-sample corrected skewness statistic uses the correction of Mardia (1970).

S3 Validity

Table S5 Fornell–Larcker criterion (diagonal = square root of AVE; below = latent correlations)

	CA	FM	ORG	LEG	PRO	POL
CA	0.775					
FM	0.500	0.706				
ORG	0.540	0.569	0.835			
LEG	0.487	0.376	0.397	0.780		
PRO	0.575	0.471	0.321	0.317	0.792	
POL	0.515	0.597	0.388	0.411	0.426	0.805

Note. Square roots of AVE are in bold.

Table S6 HTMT (below the diagonal) and HTMT2 (above the diagonal)

	CA	FM	ORG	LEG	PRO	POL
CA	–	0.508	0.541	0.501	0.567	0.512
FM	0.513	–	0.561	0.370	0.446	0.594
ORG	0.544	0.569	–	0.399	0.289	0.379
LEG	0.506	0.381	0.405	–	0.304	0.391
PRO	0.573	0.467	0.308	0.320	–	0.411
POL	0.516	0.596	0.391	0.402	0.425	–

Note. Bootstrap 95% upper limits of HTMT (2,000 resamples) range from 0.454 to 0.707, all below 0.85.

Table S7 Common method bias diagnostics

Procedure	Result	Criterion / comparison	Assessment
Harman's single-factor test	First factor explains 36.02% of variance	< 50%	Passed
One-factor CFA	$\chi^2(629) = 2,803.267$; CFI = 0.522; RMSEA = 0.136; SRMR = 0.128	Six-factor model: $\Delta\chi^2(15) = 2,077.55$, $p < 0.001$	Single factor rejected
CLF model, free method loadings	$\chi^2(577) = 650.143$; CFI = 0.984; RMSEA = 0.026; SRMR = 0.044	Six-factor model: $\Delta\chi^2(37) = 75.58$	Degenerate solution; 28.42% shared variance not interpretable
CLF model, equal method loadings	$\chi^2(613) = 723.687$; CFI = 0.976; RMSEA = 0.031; SRMR = 0.051	Six-factor model: $\Delta\chi^2(1) = 2.03$, $p = 0.154$	Fit not improved
Variance shared by the method factor (equal loadings)	12.48%	< 25%	Below benchmark

Note. The free-loading CLF solution was degenerate (variances of one or more constructs near zero), so its shared-variance estimate is not interpretable.

S4 Structural model

Table S8 Bootstrap confidence intervals (B = 2,000) and effect sizes of the structural paths

Hypothesis	Path	β	Boot. SE	95% CI of β	<i>b</i>	95% CI of <i>b</i>	f^2	Effect size
H1	CA → ORG	0.342	0.078	[0.179, 0.491]	0.360	[0.182, 0.557]	0.147	Small
H2	CA → LEG	0.390	0.087	[0.214, 0.557]	0.382	[0.200, 0.593]	0.157	Medium
H3	CA → PRO	0.452	0.082	[0.280, 0.601]	0.449	[0.275, 0.628]	0.244	Medium
H4	CA → POL	0.290	0.077	[0.137, 0.437]	0.338	[0.159, 0.529]	0.110	Small
H5	FM → ORG	0.394	0.080	[0.232, 0.544]	0.445	[0.262, 0.650]	0.196	Medium
H6	FM → LEG	0.207	0.093	[0.024, 0.385]	0.217	[0.026, 0.423]	0.044	Small
H7	FM → PRO	0.243	0.082	[0.085, 0.399]	0.259	[0.081, 0.460]	0.071	Small
H8	FM → POL	0.457	0.084	[0.294, 0.607]	0.570	[0.345, 0.863]	0.273	Medium

Note. $f^2 = \beta^2(1 - r^2)/(1 - R^2)$, $r = 0.50$.

Table S9 Bootstrap tests of the difference between the effects of control activities and financial management

Outcome	β (CA)	β (FM)	$\Delta\beta$ (CA – FM)	Boot. SE	95% CI of $\Delta\beta$	<i>p</i> (bootstrap)	Significant difference
Organizational (H1 vs. H5)	0.342	0.394	-0.052	0.146	[-0.342, 0.232]	0.697	No
Legal (H2 vs. H6)	0.390	0.207	0.183	0.167	[-0.150, 0.498]	0.253	No
Professional (H3 vs. H7)	0.452	0.243	0.209	0.151	[-0.091, 0.492]	0.166	No
Political (H4 vs. H8)	0.290	0.457	-0.167	0.148	[-0.450, 0.111]	0.268	No

Note. $\Delta\beta$ computed in each of 2,000 resamples; two-sided bootstrap *p*-values.

Table S10 Explained variance (R^2) of the endogenous constructs

Endogenous construct	R^2 (latent, SEM)	R^2 (composite, OLS)	Adjusted R^2	$F(2, 184)$	<i>p</i>
Organizational (ORG)	0.407	0.348	0.341	49.184	< 0.001
Legal (LEG)	0.275	0.233	0.225	28.019	< 0.001
Professional (PRO)	0.374	0.316	0.309	42.527	< 0.001
Political (POL)	0.426	0.326	0.318	44.416	< 0.001

Note. Composite R² from ordinary least squares on construct scores.

Table S11 Comparison of competing structural models

Model	χ^2	df	CFI	RMSE A	SRMR	AIC	BIC	$\Delta\chi^2$ vs. proposed	Δdf	p ($\Delta\chi^2$)
Proposed model (all eight paths)	733.18	620	0.975	0.031	0.052	899.2	1,167.4	–	–	–
Control activities only (FM paths fixed to zero)	789.91	624	0.964	0.038	0.072	947.9	1,203.2	56.73	4	< 0.001
Financial management only (CA paths fixed to zero)	791.23	624	0.963	0.038	0.075	949.2	1,204.5	58.05	4	< 0.001
Null model (no paths; endogenous factors uncorrelated)	1,010.70	628	0.916	0.057	0.256	1,160.7	1,403.0	277.52	8	< 0.001
Saturated model (freely correlated factors = CFA)	725.72	614	0.975	0.031	0.048	903.7	1,191.3	7.46	6	0.280

Note. AIC and BIC are computed from χ^2 ; only differences are meaningful.

Table S12 Sensitivity of the standardized path coefficients (β) across specifications

Path	Base model, ML	DWLS (Pearson)	Ordinal (polychoric) ^a	Outliers removed (N = 177)	Four correlated errors	Correlated disturbances	Method factor controlled	Demographic controls
H1 CA → ORG	0.342***	0.339***	0.318	0.314***	0.342***	0.341***	0.311***	0.346***
H2 CA → LEG	0.390***	0.398***	0.379	0.397***	0.392***	0.399***	0.308**	0.377***
H3 CA → PRO	0.452***	0.452***	0.464	0.465***	0.449***	0.452***	0.395***	0.461***
H4 CA → POL	0.290***	0.303***	0.296	0.293**	0.289***	0.289***	0.252**	0.280***
H5 FM → ORG	0.394***	0.385***	0.408	0.424***	0.395***	0.399***	0.347***	0.397***
H6 FM → LEG	0.207*	0.248**	0.257	0.221*	0.204*	0.176 ns	0.091 ns	0.213*
H7 FM → PRO	0.243**	0.217*	0.222	0.233*	0.245**	0.245**	0.178 ns	0.236**
H8 FM → POL	0.457***	0.464***	0.462	0.430***	0.459***	0.452***	0.395***	0.456***
CFI	0.975	1.000 ^b	–	0.973	0.978	0.975	0.976	0.973
RMSEA	0.031	0.000 ^b	–	0.033	0.029	0.031	0.031	0.030
SRMR	0.052	0.047	0.053	0.051	0.052	0.048	0.055	0.050

Note. *** p < 0.001, ** p < 0.01, * p < 0.05, ns = not significant. a Point estimates only. b DWLS indices truncated at their bounds.

S5 Figure

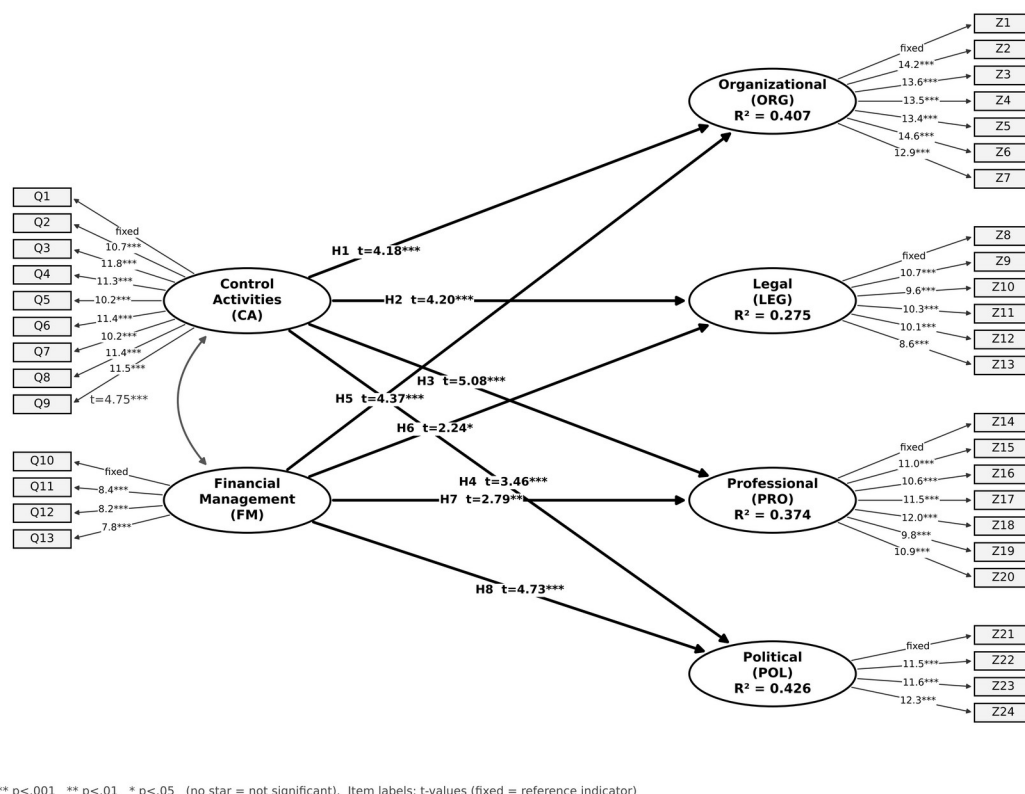


Fig. S1 Estimated structural model with t-values (fixed = reference indicator).

S6 Questionnaire

All items were rated from 1 (very little) to 5 (very much). Part A recorded gender (male, female), age (20–25, 26–30, 31–35, over 35 years), education (diploma, bachelor's degree, master's degree, Ph.D.) and professional experience (less than 5, 6–10, 11–15, over 15 years). Parts B and C were introduced as follows: “Please indicate the degree to which these financial control mechanisms are present or enforced” and “Below are questions evaluating your current perspective on the diverse dimensions of accountability within the Tax Affairs Administration.”

Table S13 Questionnaire items

Item	Construct	Question
Q1	Control activities	To what extent is there a clear segregation of roles and responsibilities in the organization?
Q2	Control activities	To what extent is each individual's duty independently reviewed by another staff member?
Q3	Control activities	To what extent is there adequate supervision provided by senior management?
Q4	Control activities	To what extent are prompt corrective actions taken to address identified weaknesses in internal controls?
Q5	Control activities	To what extent does the organization employ adequately trained personnel for implementing financial management systems?
Q6	Control activities	To what extent is a detailed and standardized chart of accounts available and utilized?
Q7	Control activities	To what extent is every organizational expenditure strictly confined within the approved budget?

Item	Construct	Question
Q8	Control activities	To what extent are budget discrepancy reports, accompanied by detailed explanations, routinely prepared?
Q9	Control activities	To what extent is there an appropriate and reliable security system for safeguarding organizational assets?
Q10	Financial management	To what extent are sufficient financial resources available to meet organizational obligations?
Q11	Financial management	To what extent are salaries and wages structured appropriately and based on professional merit?
Q12	Financial management	To what extent have unnecessary or wasteful expenditures been successfully reduced within the organization?
Q13	Financial management	To what extent have outstanding organizational invoices and liabilities been systematically settled?
Z1	Organizational accountability	To what extent are citizen visits and administrative procedures optimized for efficient service delivery?
Z2	Organizational accountability	To what extent do managers and staff consistently fulfill their professional commitments and promises?
Z3	Organizational accountability	To what extent do staff members exhibit respectful and professional behavior toward citizens?
Z4	Organizational accountability	To what extent are nepotism and administrative bribery effectively prevented in this department?
Z5	Organizational accountability	To what extent are ethical and human values adhered to in the daily behavior of the staff?
Z6	Organizational accountability	To what extent are managers and administrative staff accessible to the public?
Z7	Organizational accountability	To what extent is excessive bureaucracy (red tape) and unnecessary paperwork minimized?
Z8	Legal accountability	To what extent are statutory laws and organizational regulations correctly and strictly implemented?
Z9	Legal accountability	To what extent is prompt and accurate information regarding new laws and regulations provided to the public?
Z10	Legal accountability	To what extent are legal loopholes and citizens' regulatory grievances actively addressed?
Z11	Legal accountability	To what extent is the department's annual budget transparently presented or made accessible to the public?
Z12	Legal accountability	To what extent is the execution of overly complex laws and unnecessary legal formalities simplified?
Z13	Legal accountability	To what extent are rigorous control and supervision exercised by overarching legal authorities and the government?
Z14	Professional accountability	To what extent are citizens adequately guided by professionals to resolve their administrative issues?
Z15	Professional accountability	To what extent are mandated departmental services provided appropriately and to standard?
Z16	Professional accountability	To what extent is the current staffing level adequate relative to the operational needs of the department?
Z17	Professional accountability	To what extent is clear information regarding performance, policy execution, and departmental programs provided to citizens?
Z18	Professional accountability	To what extent do you perceive the administrative services provided to be fair and equitable?
Z19	Professional accountability	To what extent has this department succeeded in facilitating services and reducing the processing time of administrative workflows?
Z20	Professional accountability	To what extent do citizens hold trust in the commitments made by the departmental management?
Z21	Political accountability	To what extent are citizens empowered to participate in or influence the decision-making processes of this department?
Z22	Political accountability	To what extent is public or stakeholder feedback integrated into the selection and evaluation of departmental managers?

Item	Construct	Question
Z23	Political accountability	To what extent has this department succeeded in eliminating administrative discrimination and ensuring equal societal treatment?
Z24	Political accountability	To what extent is there a clear alignment between the department's executive programs and the broader political expectations of society?

Note. Control activities and financial management items were adapted from Wakiriba et al. (2014). Accountability items were adapted from Mulgan (2000), Bovens (2007) and Romzek and Dubnick (1987).